Specify the Model

1.0 Overview

1. What is the purpose of the model and is there a research question affiliated with this model?

The purpose of this model is to find how and colony survival will be affected by poisonous food sources. Specifically, what is the most ideal amount and dose of poison in order to exterminate the colony while ensuring ants do not die out before poison transmission.

2. For whom is the model designed? What kinds of entities are in the model?

This model is designed for anyone who wants to research insect behavior and insect colony survivability. My model focuses on toxins and ant colony extermination, but professions like exterminators could find this information useful. The main entities in this model are the ant agents, food sources, and the central nest / colony.

3. By what attributes for example state variables and parameters are these entities characterized?

Ants are characterized by variables such as health, movement speed, and searching state. Food sources are characterized by location from the nest, toxicity level, and amount / size. The nest is characterized by its population size, and overall poison level.

4. What are the exogenous factors slash drivers of the model?

The exogenous factors are the food density and the amount of toxin inside the food sources. The model will set these factors at the beginning of the simulation to then observe how these factors will affect the agents.

5. If applicable, what are the temporal and spatial extents of the model?

The spatial extents of this model are a 2D lattice / grid. The model will allow ants to move on a 2D plane and deposit their pheromones on this 2D plane. The temporal aspect of this model will have ants making decisions on each tick of the model. These ticks could be increased or decreased in order to speed up or slow down ant movement and decision making.

6. What entity does what and in what order?

   The first entity to react are the pheromone trails. They will decay first, then the ant agents will process the environment. Each ant agent will move in a direction based on searching state and nearby pheromone trail strength. Finally, the nest will update its poison level based on any new food delivered from returning ant agents.

2.0 Design concepts

1. What general concepts, theories, or hypotheses are underlying the models' design?

   The general concept being explored here is that ants will be able to tell one another where to move based on depositing a trail of pheromones. The model is assuming that ants do not have any other form of communication or trail types based on danger or other circumstances. The model's hypothesis is that the more poisonous food delivered to the nest will result in the total ant population decreasing. This will be determined by a threshold limit based on the ant population size.

2. On what assumptions are the agents' decision models based?

   The model assumes that the ant agents will prefer to move towards stronger pheromone trails while searching for food over any other factors. The model also assumes that ant health will be negatively impacted once grabbing a poisonous food source. Furthermore, the model assumes that ant movement speed will decrease with diminishing ant health.

3. Why are certain decisions chosen?

   The model chooses to have ant speed decrease with diminishing health to show some sort of negative physical effect on the ant agents. Without this effect, there would be less visible recognition in the model for when an ant consumes a poisonous food source. We could remove this factor and just have ant health decrease until the ant agent dies, however decreasing movement speed can replicate and show visually negative physical effects of consuming a poisonous food source.

   Furthermore, we assume ant agents want to move towards strong pheromone trails in order to reinforce ant trail formation. The model wants to assume ants can find an

efficient way to travel to and transport these food sources, and focus on the effects of food toxicity on these trails.

4. If the model is based on empirical data, where do the data come from?

The model is not based on empirical data and is only based on biological observations of ant foraging patterns. The model aims to see how these biological patterns change with the presence of poisonous food sources, rather than specifically model exact ant trail formation.

3.0 Agents

1. Heterogeneity/Homogeneity? Different subgroups? Appearance?

   The agents in this model are all homogeneous. All ant agents will have the same ruleset and start with the same state. Ants will also all have the same attributes like start health and start movement speed. There are no subgroups of agents defined in this model. We aim to reduce complexity by treating each ant the same. This model could be expanded in the future by adding more ant variants and randomizing poison resistance among ant agents.

2. Deterministic vs. random behavior

Ant agents will have random movement during the searching phase to mimic natural ant exploration. This random movement will be effected only if there is a pheromone trail present. We assume that ant agents will know the path back to the nest after finding a food source, so this movement while returning will be deterministic. This allows each ant agent to explore in random patterns while searching for a food source, and then allow them to release a pheromone trail while they move back to the nest decaying over time.

3. Which behavior should agents show?

Ant agents should show random searching behavior. The model should visually depict ants moving in random directions while in a searching state. The ant agents should also show trail formation behaviors by preferring to move in the direction of a pheromone location if one is detected near the ant. This should result in visual trail formation in the model. Finally, ant agents should be able to show returning behavior to the nest. The model should visually depict ants moving straight back to the nest after finding a food source.

4. Define (sets or subsets of) attributes (for defined state)

Ant agents will have the following attributes. Health, XY position, movement speed, and searching state. This searching state will determine how the ant moves in each tick. The states are searching or returning. An ant with the searching state will have random movement while an ant with a returning state will have deterministic movement back to the ant nest.

5. Define interactions between agents

Ant agent interactions are not direct, but indirect through pheromone trails. An ant that has found a food source and is returning will deposit a pheromone trial at its location decaying over time. This trail will then affect any other ant agent that comes near it, incentivising the ant to move in that direction. Pheromone trails will be the main interaction between agents in the model.

4.0 Environment

1. Which information is given to the agents?

The only information given to the ant agents is the information present in their direct surroundings. This would be considered the cells above, below, and to the left and right of the ant agent. The information about the environment that the ant agent would get would be if a pheromone trail was present, if there was a food source present, or if they have reached the nest / colony.

2. How does it influence the actions of agents?

This information influences the agents by mostly just determining agent movement. By detecting a food source / pheromone trail/ nest, this will determine where they move next. In all cases they will prefer to move towards these targets in the environment, else they will move randomly.

5.0 Requirements

1. Identify emergent aspect/Identify a kind of dynamic

The emergent aspect of this model is how different number of trails and the foraging strength of these trails lead to the extermination of an ant colony. The

model would show this through nest toxicity spiking and increasing rapidly. Another emergent behavior would be trails breaking apart. This would be the result of food toxicity levels being too high, resulting in ants moving too slow or dying before enforcing trail patterns.

2. Decide for a way to design agent-based models

   This model would be designed by first focusing on individual ant rules in a bottom up approach. These individual rules would then be modeled together with multiple ant agents to see trends in the ant colony. Furthermore, environmental variables such as the amount of food / location of food sources / toxicity levels should be able to be altered before the model runs in order to see their effects.

3. Choose processes and adaptive key processes

   The main process is the ant foraging loop where ants will grab food and return it to the nest. Adaptive key processes will be how ant speed and health change as they ingest and carry poisonous food sources.

4. Think of contexts well-suited for agent-based models

   Contexts well suited for agent based models include contexts where individual behavior can be modeled together as a part whole relationship. These individual behaviors should be able to build off of each other between agents to see emergent behaviors. The ant model for poisonous food sources is a great example context. Ants act as our individual agents with their own rule sets. When modeled together you can see the agents interact with each other through trail formation in order to deliver poisonous food to the colony.

5. Define entities, state variables, and scales

   The entities in this model include the ant agents, pheromone trails, food sources, and the colony / nest. The state variables for the ant agent include, ant health, is carrying food, search state, movement speed. The state variables for food sources include, poison level, and amount of food. The state variables for the nest is the total poison delivered and the ant population count. The state variable for pheromone trails would be the decay rate. The spatial scale for the model includes a grid or 2D lattice and the time scale of this model would be modeled by ticks.

6. Design concepts of model

The design concepts of this model rely on agent coordination through the environment. Ant agents will not directly interact with each other, only influence the environment around them and in turn affect other agents. Another important design choice is having agents only have information on themselves and the direct environment around them. No ant agent should have any other information on food locations, toxicity levels, or other agents. Furthermore, stochasticity is another important design concept in this model. Ant agents should move randomly in order to mimic natural foraging.

7. Define initial state of agent-based model

   The initial state of this model includes the starting number of ants in the colony, the ants maximum health / movement speed, food locations with their corresponding toxicity levels, and a clear 2D lattice with no pheromone locations. This ensures that the model starts fresh and can build off itself once the ants begin to forage.

8. Define model dynamics

   The model dynamics are shown through the ant mortality rate and the pheromone trail formation. As time progresses, ants will begin finding food sources creating a pheromone trail as they move. This will allow the model to start to reinforce trails for other foraging ant agents. As the model progresses, ant agents will begin losing health as they come in contact with poisonous food sources. The model will then show ants slowly dying as they continue to stay in contact with these food sources.

6.0 Topology

1. What should it represent? Physical/geographical/social… network

The topology of this model represents a 2D grid or lattice environment. Each grid or cell can hold an ant agent, food source, pheromone location, or nest location in it. This could be used to model how ants move across a floor in a house, or a flat surface in the environment. This model does not take any 3D movement into account.

2. Is this static or dynamic?

The actual 2D grid environment is static, however the pheromone trails will be able to layer on top of this grid forming dynamic pheromone trails. Every tick of the

model ant agents can add pheromone trails at their location. Furthermore, present pheromone trails will decay after each tick.

3. Define number of topologies (1 or more)

There is one main topology which is the 2D grid environment. There is also the pheromone trail topology that is layered on top of the grid as time progresses.

7.0 Decision making

1. What are the subjects and objects of the decision making?

The subjects of decision making include the ant agents. They make decisions on where to move each tick. The objects of decision making include the nest, food sources, and pheromone trails. Each object influences where the ants decide to move. If an ant agent senses a food source nearby, it will move towards it until it can pick up the food source and return to the nest. The nest will influence how the ants move when returning with a food source based on the nest location. The pheromone trail will also influence how the ant moves when searching for food.

2. Are multiple levels of decision making included?

The model does not have multiple levels of decision making. The only decision making being made are by the individual ant agents.

3. What is the basic rationality behind agents' decision making in the model?

The rationality behind the agents decision making are incentivised items in the environment. An ant agent will prefer to move towards a pheromone trail or food source if one is present around the agent. An ant agent should also know the nest location once leaving and will move directly towards it after finding a food source.

4. How do agents make their decision, what is it based on?

The agents make their decision based on the direct items in the environment around them. If an ant agent has a food source, or pheromone trail around them they will then prefer to move in that direction.

5. Do social or cultural values play a role in the decision-making process?

No social or cultural roles play a value in the decision making process. The decisions are solely based on biological incentives.

6. Do spatial aspects play a role in the decision process?

Yes, special aspects are extremely important to the ants' decisions involving pheromone trials. This is because the further away a food source is, the more time it will take an ant to return to the nest. In turn, this means the pheromone trail will decay more as the ant returns. This results in ants' decision to follow pheromone trails to be harder as less trails will be present.

7. Do temporal aspects play a role in the decision process?

Temporal aspects do play a role in the decision process. The pheromone trails decay over time meaning the more time passes the less information each ant agent has on where a possible food source might be.

8. To what extent is uncertainty included in the agent’s decision rules?

Uncertainty is included in the ants' searching patterns. The ant agents will move in random directions while searching allowing random foraging patterns to emerge.

9. Do the agents learn?

Agents do not learn individually. Rather, agents will reinforce pheromone tails over time allowing other agents to follow the trails to get to other food sources easier.

10. Does the environment learn?

The environment does learn through the pheromone trail formation. These trails allow the environment to create a pathway for other agents to access food sources easier without traveling randomly.

11. What state variables are assumed for agents in the sensing process?

The state variables assumed for sensing the environment include being able to determine what is located in the agent's neighboring cells. This could be a food source, pheromone trail location, or the ant nest.

12. Is the sensing ever erroneous?

    Sensing is not erroneous. The agents are assumed to have perfect sensing through the foraging process being able to determine with 100 percent confidence what is located in their neighboring cells.

13. Does the environment sense?

The environment does not sense anything. The environment can only be added to by ant agents releasing pheromones.

8.0 Agent prediction

1. Are the interactions among agents and entities direct or indirect?

   The interactions between agents are indirect. These interactions are modeled through the pheromone trail formation.

2. Might agents be erroneous in the prediction process and how is this implemented?

Yes agents can be erroneous in the prediction process. The agents could follow a pheromone trail to a depleted food source. This could possibly happen if an ant agent follows the pheromone tail quick enough after a food source has been consumed.

9.0 Interaction

1. On what do the interactions depend?

The interaction depends on the pheromone strength. This strength determines how much each ant agent deposits when it moves away from a food source. A higher pheromone strength will result in stronger trail formation.

2. If the interactions involve communication how is the communication represented?

The interactions do not involve any communication between agents. All interactions are independent.

3. On what do the interactions depend?

The interaction depends on the pheromone strength. This strength determines how much each ant agent deposits when it moves away from a food source. A higher pheromone strength will result in stronger trail formation.

10.0 Collectives

1. Do agents form or belong to a collective body?

Yes, the agents all belong to a collective colony. This colony is able to form visual trails while foraging for food.

2. How is this collective body represented?

The colony is represented through the ant nest. The ant population number and nest poison level are monitored through the nest's attributes. The colony is also able to be seen visually through trail formation in the model.

3. Does the collective body act for or instead of the agents?

The collective body does not act for or instead of the agents. The ant colony is only representative of the ant population and total ant poison threshold.

11.0 Stochasticity

1. What processes are modeled by randomness?

The ant movement when searching is random. This allows ant agents to mimic natural foraging patterns when looking for food sources.

2. What agents are modeled by randomness?

Only the ant agents are modeled by randomness. This randomness in models when they are searching for food.

12.0 Simulation

1. How is the model seeded? What are the initial states?

The model is seeded with a start count of 100 ants. The initial states depend on the food source locations and amounts placed by the user. Furthermore, the nest position level should be set to 0 for the initial state.

2. How does the simulation run, for how many cycles or how much time?

The simulation will run until the ant population reaches zero or until the user stops the simulation. There is no set cycle time for the simulation run time.

3. What do you expect to happen during the simulation?

During the simulation I expect to see ants moving in random patterns for the first 100 cycles. I expect to see a few ants start to find food sources randomly and begin returning to the nest. As the simulation continues, I expect to see ant trails begin to form the pheromone trails. I also expect to see the nest poison level increase throughout the simulation.

4. What is the expected end state of the simulation?

The expected end state of the simulation is for every ant to be exterminated or for the nest poison level to reach a critical threshold. This will represent ant colony extermination.

5. What result do you expect?

I expect to find an optimal amount of food with an optimal amount of food toxicity for exterminating the ant colony with the least amount of cycles. I expect there to be a poison tipping point, where too much poison will result in ants not being able to make it back to the ant nest.

6. How do you expect to convey the result to sponsors? (graphs, probability, etc.)

I would convey the results using graphs. I would do a graph for different initial states. For example a graph with 1, 2, 3, or more food sources. I would also have a graph for different toxicity levels. For example low, medium, high. Then I would have the graphs model ant population vs time. This would be able to clearly demonstrate how the model works with different inputs.

7. What problems do you anticipate with the simulation?

One problem I anticipate is ants dying off too quickly. The total ant population could die before even reaching the nest resulting in the model not working as intended.

8. How would you address those problems?

To address these problems I would make sure the poison level is not too high. I would have to do this through testing the model and adjust different parameters such as ant health and movement speed.

13.0 Logistics

1. If you were to program this model, what would you use (i. e. NetLogo, Python, etc.)?

If I were to program this I would use C++ with Raylib for the graphics. I have experience using both. Furthermore, the model is not too complex and could be modeled using simple graphics and inputs.

2. How would you present the result of the simulation? (i. e., A paper, a demonstration, Github repository, etc.)

I represent the model through a readme and a Github repository. I would have a paper (similar to this) along with some graphs and instructions on how to run the model.

14.0 Other

1. Are there other aspects to the model that you want to include here?

One aspect of the model not talked about would be the possibility to add food sources while the simulation is running. You possibly could see interesting behaviors when adding food sources close to enforced trails.

2. Are there other notes that you want to remind yourself about if you would choose to build this model?

I would like to remind myself about tweaking pheromone trail decay rate and detection for ant agents. I need to make sure that the agents do not follow any slight pheromone location and that there is a strong enough re-enforced trail from multiple other agents before an ant will consider moving towards a pheromone location. I could use probability as well, where if an ant detects a slight pheromone trail, it is more likely to move towards it, but will not always move in that direction.

3. Will you be building the model in NetLogo or writing the model in pseudocode?

   I will be doing pseudo code for this model. I plan to code this model in C++ over the sumer and display it as a project on my Github.

Pseudocode

```
// Initialize variables
Initialize food strength
Initialize average food distance from nest
Initialize food density
Initialize ant speed
Initialize ant population
Initialize pheromone trail decay rate
Initialize nest position level to 0

While ant population > 0
        // Decrease all active pheromone trailers
        For each pheromone location
                Decrease Pheromone strength by decay rate

        // Update each ant
        For each ant
                Calculate speed based on health
                If ant state is searching
                        If pheromone is detected
                                If low strength
                                        20% chance ant will move toward pheromone
                                If medium strength
                                        50% chance ant will move toward pheromone
                                If high strength
                                        90% chance ant will move toward pheromone
                        else if food source is found
                                Change ant state to returning to nest
                                Begin releasing pheromone trail
                        else
                                Move randomly
                If ant state is returning to nest
                        If not at nest
                                Move towards nest
                                Decrease pheromone strength
                                Decrease health depending on poison strength
```

```
        If ant health < 0
                Remove ant from population
else if at nest
        Deposit Food and increase nest poison level
        Change ant state to searching
        Stop releasing pheromones
```

Expected Behavior

I expect after running this pseudo code multiple times, I will see emergent behavior based on the inputs. I expect that if the poison strength and food amount is low, for the ant colony survival rate to remain steady with the chance of the ant colony not being exterminated. I also expect lower trail formation with less food sources. If the poison strength is high, along with the number of food sources, I expect that ant trails will break apart as ants die before reinforcing them. I also expect the pheromone strength ants emit to determine heavily how fast trails are formed and re-enforced. I expect to find a medium amount of toxicity level of food where ant trails can be reinforced while delivering a steady stream of poison back to the nest. This should result in the optimal time to exterminate the ant colony.